

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy trends across primary energy and electricity generation

Source: Our World in Data – Renewable Energy | Authors: Hannah Ritchie, Max Roser, Pablo Rosado

75% of Global GHG Emissions Still Come from Fossil Fuels

- Three-quarters of global greenhouse gas emissions come from burning fossil fuels for energy.
- Fossil fuel combustion causes at least 5 million premature deaths per year from local air pollution.
- Since the Industrial Revolution, most countries' energy systems have become dominated by coal, oil, and gas.
- To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon sources — nuclear and renewable technologies.

The Urgency

Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources. Traditional biomass is typically excluded from modern renewable statistics.

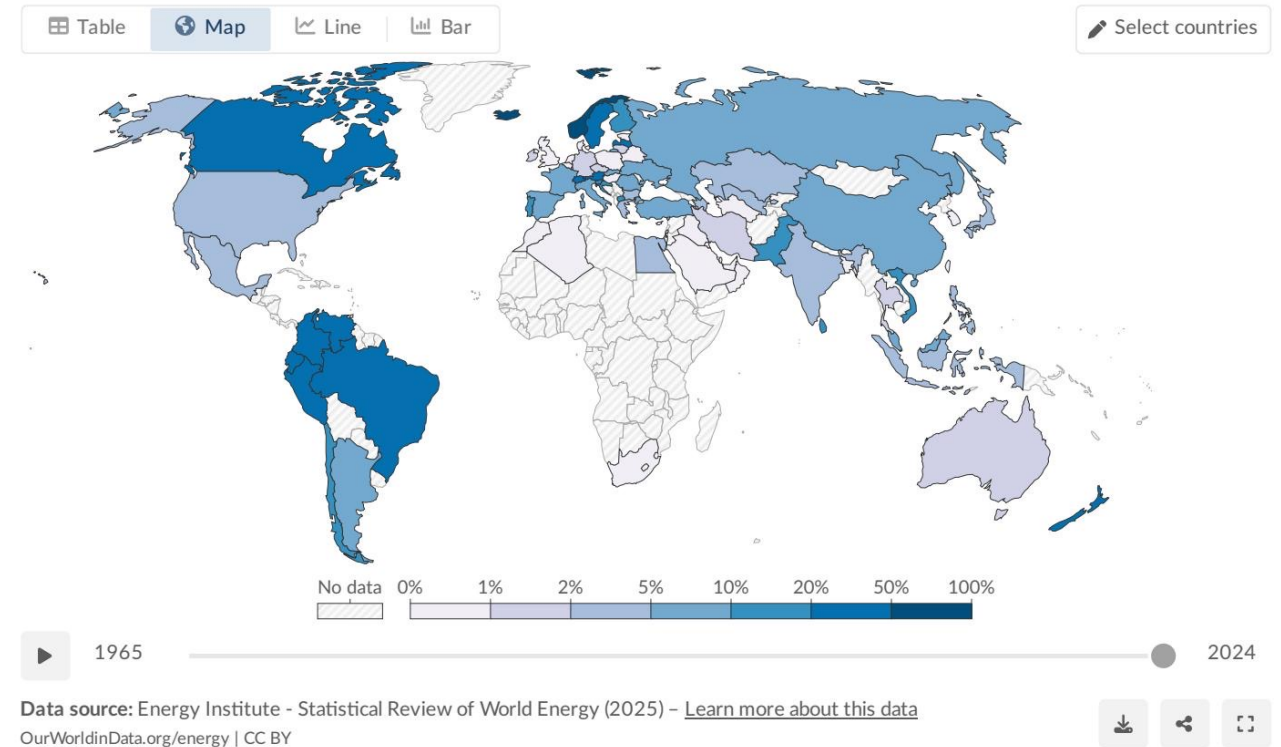
Renewables Now Supply ~14% of Global Primary Energy

Approximately one-seventh (~14%) of the world's primary energy now comes from renewable technologies (2024 data, substitution method).

Primary energy encompasses electricity, transport, and heating combined.

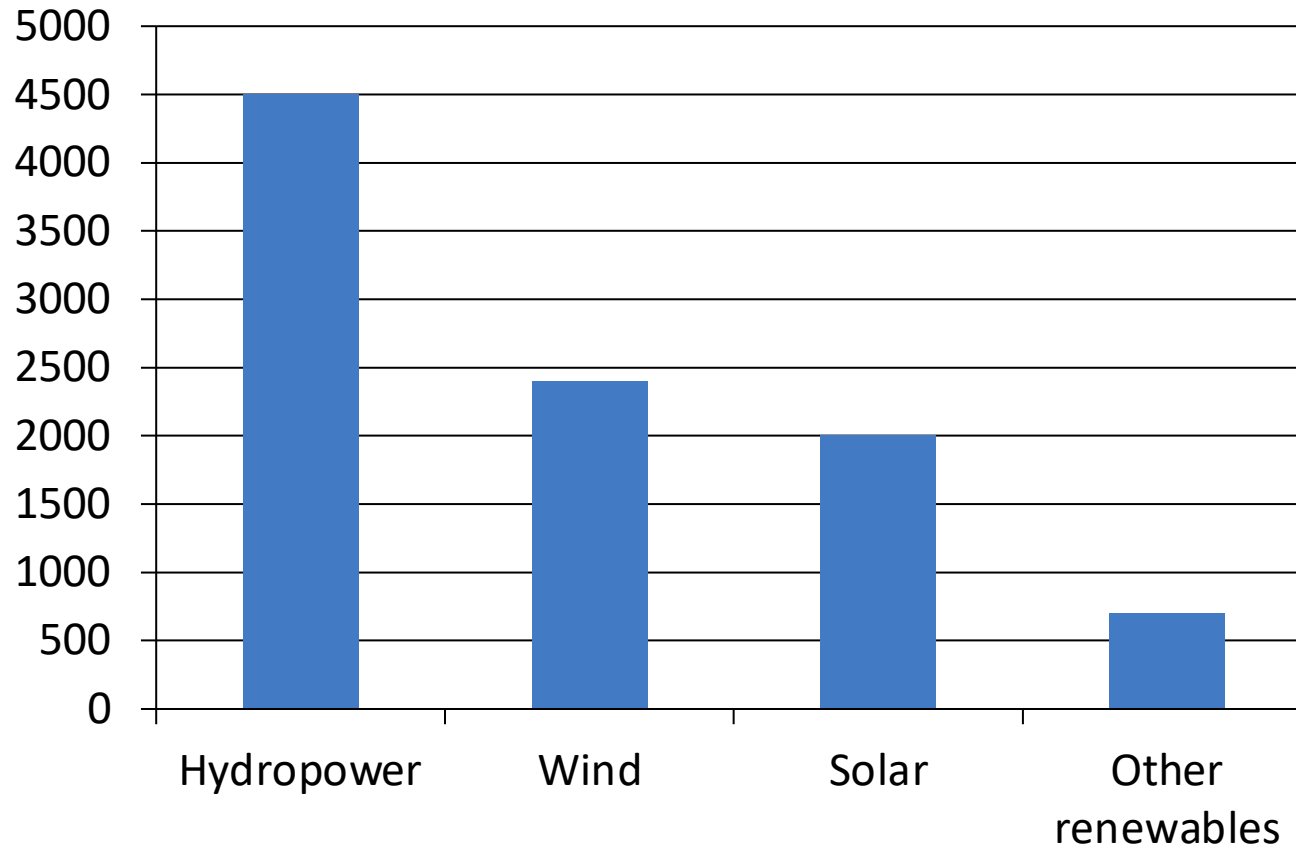
Because transport and heating are harder to decarbonise (more reliant on oil and gas), the renewable share in total energy is lower than in electricity alone.

Regional variation is significant: Norway, Brazil, and New Zealand derive large shares from renewables, while fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global Renewable Electricity Generation by Technology (2024, TWh)



- Hydropower remains the single largest source at ~4,500 TWh (~47% of renewable electricity).
- Wind has grown rapidly to ~2,400 TWh (~25%).
- Solar is the fastest-growing source at ~2,000 TWh (~21%), with dramatic growth since 2010 driven by falling costs and policy support.
- Other renewables (geothermal, biomass, wave, tidal) contribute ~700 TWh (~7%).

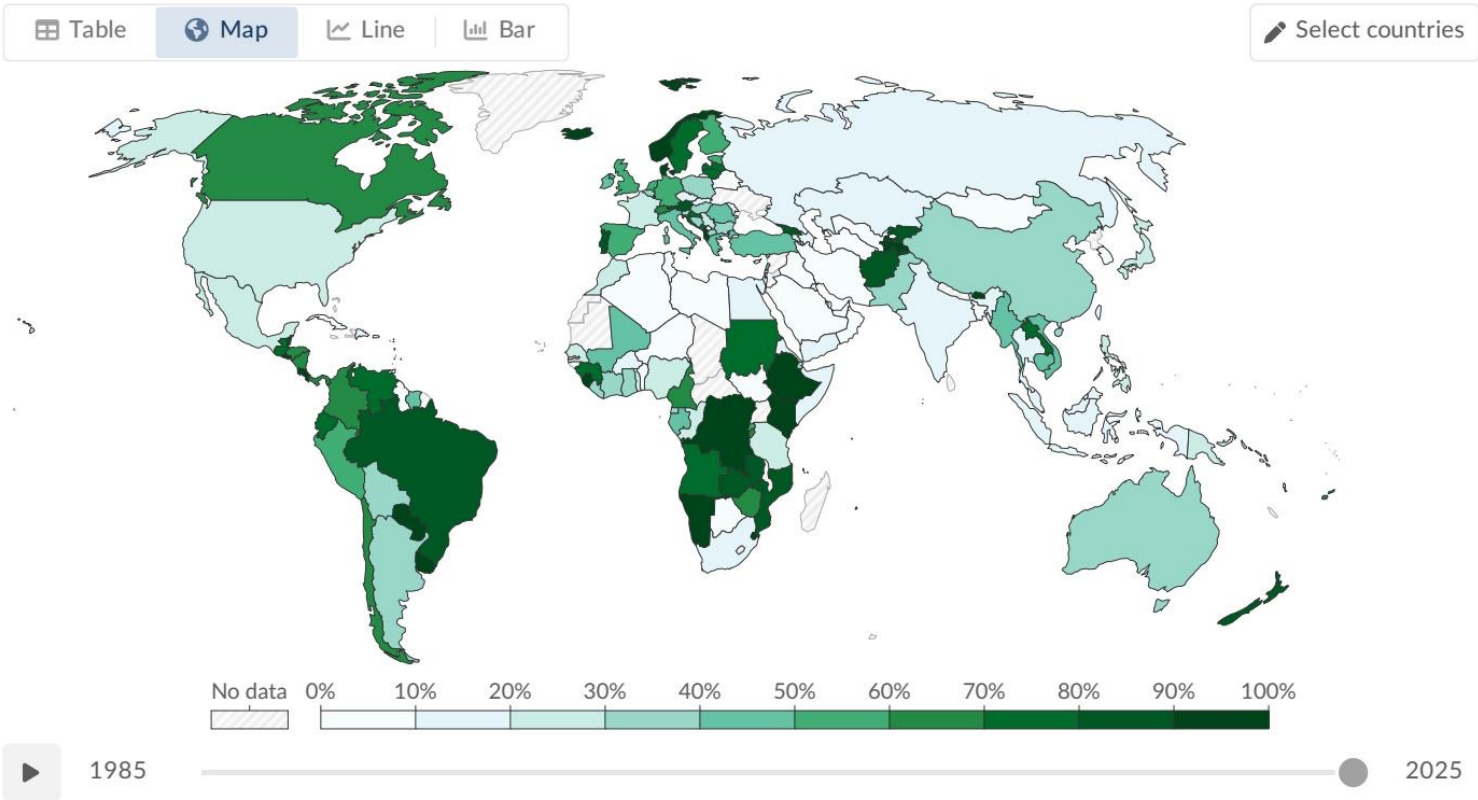
Wind and solar combined now nearly match hydropower's output and are growing much faster.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
United States	~25%
India	~25%
Saudi Arabia	<2%

World average: ~30%
Several African and Middle Eastern countries remain below 10%.

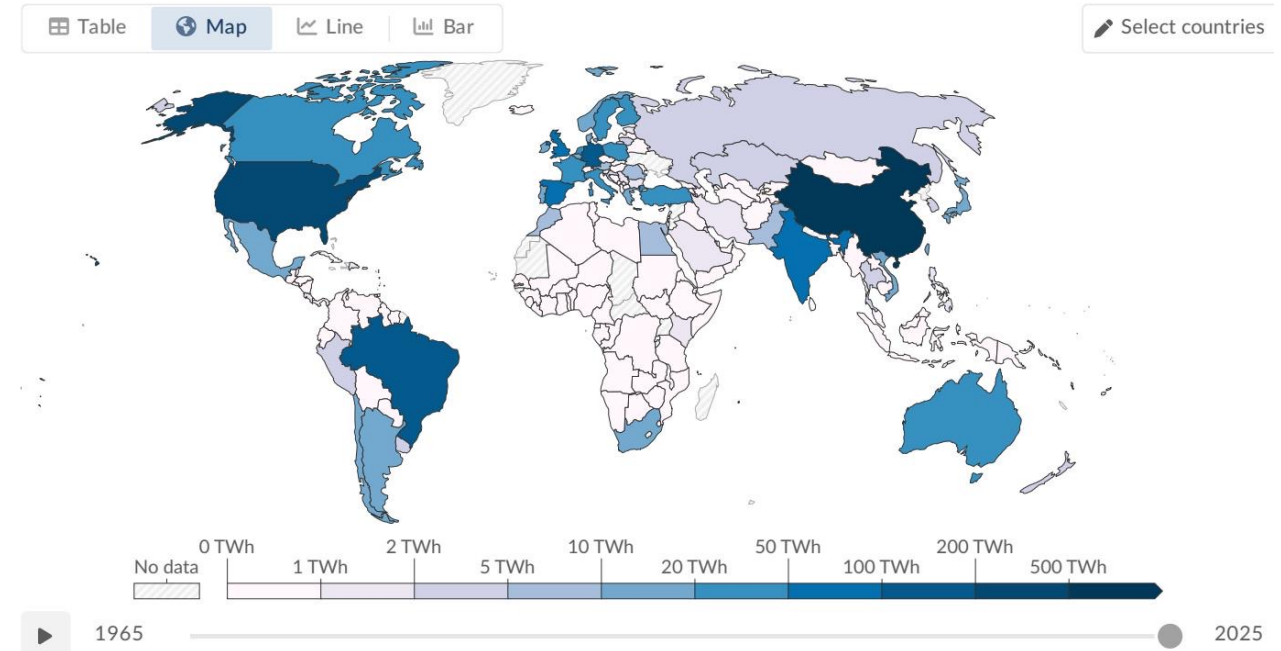
Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower accounts for roughly half of all renewable electricity generation (~4,500 TWh in 2024).

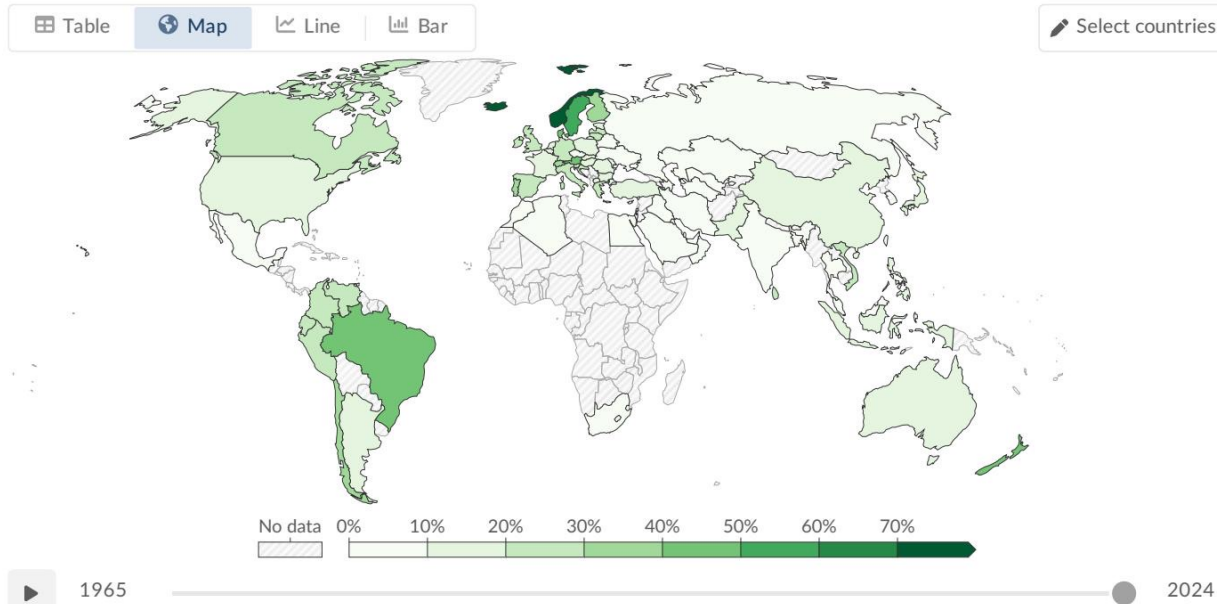
It has been generating electricity at scale for over a century.

Top producers include China, Brazil, and Canada.

While hydropower still leads, its growth potential is limited compared to wind and solar.



Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind vs. Solar Comparison (2024)

Wind Energy

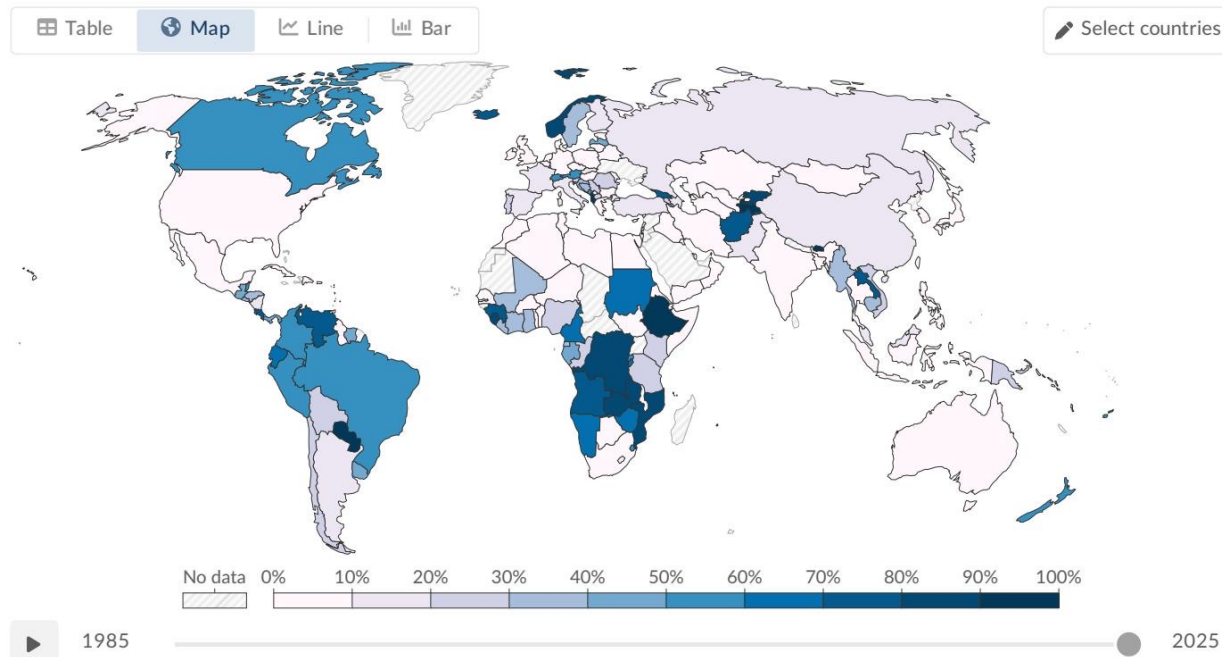
- Generation: ~2,400 TWh
- Share of renewable electricity: ~25%
- Major markets: Europe, China, United States
- Includes onshore and offshore installations

Solar Energy

- Generation: ~2,000 TWh
- Share of renewable electricity: ~21%
- Fastest-growing source with dramatic growth since 2010
- Driven by falling costs and policy support

Combined, wind + solar (~4,400 TWh) now nearly match hydropower (~4,500 TWh).

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



The transition to renewables is highly uneven across regions.

Leaders:

- Scandinavia, Brazil, and Canada benefit from abundant hydropower (65–98% renewable electricity).
- European nations like Denmark and Germany have invested heavily in wind and solar.

Laggards:

- Much of Sub-Saharan Africa has minimal renewable electricity infrastructure.
- Many Middle Eastern nations remain almost entirely dependent on fossil fuels for electricity (shares below 10%).

These disparities reflect differences in natural resources, investment capacity, and policy priorities.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14% of primary energy, ~30% of electricity

Renewables have reached significant milestones, but shares remain far below what is needed for climate targets.

Hydropower still dominates

Generating roughly half of all renewable electricity, yet its growth potential is limited compared to wind and solar.

Wind and solar are the fastest-growing technologies in history

Combined, they now produce nearly as much electricity as hydropower and are expanding rapidly.

Stark regional disparities persist

Scandinavia and Latin America exceed 60–98% renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.

Transport and heating lag far behind electricity

The harder-to-decarbonise sectors keep overall primary energy share much lower than the electricity share.

The Path Forward

To meet climate targets, the pace of renewable deployment must accelerate dramatically — especially in transport, heating, and regions currently left behind.